



Department of Education

STEM Education

National Schools of Excellence STEM Chemistry Teacher's Guide Grades 11 Term 2

<u>DOMAIN</u>	<u>Phylum</u>
SCIENCE	Chemistry
Technology	
Engineering	
Mathematics	

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ISBN

Acknowledgements

This STEM Chemistry Teacher Guide for Grade 11 constitutes ten (10) Thematic Areas (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented.

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

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SECRETARY'S MESSAGE

This Chemistry Teacher's Guide is to be used exclusively by teachers of Chemistry to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Chemistry Bridging Teacher's Guide ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Chemistry before undertaking the advanced concepts introduced in this Teacher's Guide.

The Teacher's Guide conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Chemistry as a fundamental field of science.

This STEM Chemistry Teacher's Guide for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three (3) Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Chemistry.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Chemistry is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Chemistry to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this Teacher's Guide is accompanied by a syllabus and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this Teacher's Guide as the official curriculum for Chemistry to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Chemistry is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Chemistry help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Chemistry is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Chemistry.

The Grade 11 Chemistry Teacher's Guide entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Chemistry is a highly specialized subject students undertaking Chemistry are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Teacher's Guide are sequenced to guide teachers of Chemistry through the process of imparting basic chemistry knowledge and skills to students.

THEMA	MODULE	TOPICS
BIODIVERSITY	Ethnobotany	1. Medicinal/Herbal Plants 2. Bioactive compounds in plants 3. Examples of Therapeutic Drugs from plants
BIOTECHNOLOGY	DNA Analysis and Proteins	1. Polymerase Chain Reaction 2. Dyes in DNA Sequencing 3. Proteins
HEALTH AND MEDICINE	Biochemistry in Medicine	1. How a medical drug works (mechanism of action) 2. Drug Pharmacology 3. Significance of Diagnostic Laboratory Tests
ENERGY		
BATTERY TECHNOLOGY		
INFORMATION SYSTEM		
EARTH SCIENCE		
WASTE MANAGEMENT		
INFRASTRUCTURE		
COMPUTER SCIENCE		
ARTIFICIAL INTELLIGENCE		
ENTERPRENEURSHIP		

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honorable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is a public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Chemistry Bridging Teacher Guide students were introduced to the basic concepts and knowledge of Chemistry in order for them to develop an appreciation of the discipline in a broader context. In this Teacher Guide students are introduced to advanced concepts in Chemistry as they are applied to the Ten (10) Thema.

The introduction of the STEM Curriculum, especially the Bridging Program and the Grades 11 and 12 Syllabi, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

STEM DOMAIN

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

It comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

It concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM CHEMISTRY CONTENT OVERVIEW – TERM 2

Table 1.1: Thema Sequence and Duration

Grade 11 Term 2: STEM-proper Course		
BIODIVERSITY (3 Weeks)	Ethnobotany	1. Medicinal/Herbal Plants 2. Bioactive Compounds in Plants Examples of Therapeutic Drugs from nature
BIOTECHNOLOGY (3 Weeks)	DNA Analysis and Proteins	1. Polymerase Chain Reaction 2. Dyes in DNA Sequencing 3. Proteins
HEALTH AND MEDICINE (3 Weeks)	Biochemistry in Medicine	1. How a Medical Drug Works (Mechanism of Action) 2. Drug Pharmacology 3. Significance of Diagnostic Laboratory Tests

Table 1.2 Sequencing & Benchmarking STEM Chemistry Content Standards

Setting STEM Chemistry Content Standards for Progressive Learning & Assessment			
Thema	Biodiversity	Broad Content Standard	
Module 1	Ethnobotany	11.1a: Scientifically document medicinal plants in Papua New Guinea 11.1b: Discuss bioactive compounds and biomolecules from plants 11.1c: Identify and discuss therapeutic drugs from nature	
Topics	Performance Indicators	Module Benchmark	Assessment Tasks
1.Medicinal/Herbal Plants	11.1.1: Scientifically document medicinal plants in Papua New Guinea	Students' ability to demonstrate knowledge and skills in; - Medicinal plants documentation in PNG - Important bioactive compounds and biomolecules from plants - Therapeutic drugs derived from nature	Assignment + Written Test
2. Bioactive Compounds in Plants	11.1.2a: Define bioactive compounds 11.1.2b: Describe the uses of bioactive compounds in plants 11.1.2c: Discuss different techniques for separation of biomolecules from plant extract 11.1.2d: Test crude plant extracts against a target 11.1.2e: Observe their effects against the target		
3. Examples of Therapeutic Drugs from Nature	11.1.3a: Identify therapeutic drugs from nature 11.1.3b: Discuss the importance of the therapeutic drugs.		
Thema	Biotechnology	Broad Content Standard	
Module 1	DNA Analysis and Proteins	11.1 Discuss DNA Analysis in Polymerase Chain Reaction, Dyes DNA Sequencing and Proteins	
Topics	Performance Indicators	Module Benchmark	Assessment

	Students are expected to:		Tasks
1. Polymerase Chain Reaction	11.1.1: Describe the components, procedures and applications of polymerase chain reaction	Students' ability to demonstrate knowledge and skills in; • Components, procedures and applications of polymerase chain reaction in health and medicine • Dyes In DNA sequencing • Proteins as important macromolecule in living functions.	Assignment + Written Test
2. Dyes in DNA sequencing	11.1.2a: Discuss the importance of dye chemistry in DNA sequencing 11.1.2b: Discuss different sequencing procedures		
3. Proteins	11.1.3a: Define proteins 11.1.3b: Identify types of proteins 11.1.3c: Discuss synthesis of proteins 11.1.3d: Discuss case studies of proteins utilized in biotechnology		
Thema	Health and Medicine	Broad Content Standard	
Module 1	Biochemistry in Medicine	11.1a: Discuss types of drug design, 11.1b: Expound on how drugs work 11.1c: Signify diagnostic laboratory results	
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks
1. How a Medical Drug Works (Mechanism of Action)	11.1.1: Describe how a medical drug works	Students' ability to demonstrate knowledge and skills in; • How a medical drug works • Drug pharmacology • Significance of Diagnostic laboratory results	Assignment + Written Test
2. Drug Pharmacology	11.1.2: Describe clearly drug pharmacology		
3. Significance of Diagnostic laboratory tests	11.1.3a: Discuss the types of diagnostic laboratory tests. 11.1.3b: Describe the significance of diagnostic laboratory tests.		

Table 1.3 Mapping Content Progressions by Modules

Thema/Module	Broad Content Standard	1	1	1
Biodiversity/ 1. Ethnobotany	11.1a: Scientifically document medicinal plants in Papua New Guinea 11.1b: Discuss bioactive compounds and biomolecules from plants 11.1c: Identify and discuss therapeutic drugs from nature	✓		
Biotechnology/ 1. DNA Analysis and Proteins	11.1 Discuss DNA Analysis in DNA Sequencing, Dyes in DNA sequencing and Proteins		✓	
Health and Medicine/ 1. Biochemistry in Medicine	11.1a: Describe how a medical drug works 11.1b: Expound on Drug pharmacology 11.1c: Signify diagnostic laboratory results			✓

THEMA

BIODIVERSITY

INTRODUCTION

Biodiversity is short for biological diversity. It includes all organisms, species, and populations; the genetic variation among these; and all their complex assemblages of communities and ecosystems. It also refers to the interrelatedness of genes, species, and ecosystems and their interactions with the environment. Biodiversity is a key measure of the health of any ecosystem, and of our entire planet. Every organism in an ecosystem, or biome, relies on other organisms and the physical environment. For example, plant and animal species need each other for food, and depend on the environment for water and shelter. Biodiversity describes how much variety an ecosystem has, in terms of resources and species and also genetically within species. A more diverse ecosystem will have more resources to help it recover from famine, drought, disease or even the extinction of a species. There are several levels of biodiversity, each indicating how diverse the genes, species and resources are in a region.

In this Chemistry Phylum, the focus is on ethnobotany in an effort to introduce students to the knowledge of the traditional uses of botany (plants) for different purposes but with specific emphasis on medicinal uses or herbal products.

Module 1: Ethnobotany

3 weeks

Native people all over the world have traditional uses of plants for various purposes, such as for medicines, foods, intoxicants, clothing, building, and so on. The scientific study of the relationships between native people and the traditional uses of the plants is known as Ethnobotany. Ethnobotanists are the scientists that undertake these studies and document the existing relationships for proper scientific purposes.

In this module, the students will be required to list plants that are of use to their traditional settings and properly document them for further scientific studies. The emphasis is on the traditional uses of plants for medical purposes.

Linking Previous Module (STEM Bridging)

Pre-requisite:

- Module 11.1: Properties and structure of matter
- Module 11.2: Types of Chemical Reactions

Broad Content Standard

11.1a: Scientifically document medicinal plants in Papua New Guinea

11.1b: Discuss bioactive compounds and biomolecules from plants

11.1c: Identify and discuss therapeutic drugs from nature

Content Benchmark

Students' ability to demonstrate knowledge and skills in;

- Medicinal plant documentation in PNG
- Important bioactive compounds and biomolecules from plants
- Therapeutic drugs derived from nature

Topic 1: Medicinal/ Herbal Plants

Performance Indicators;

Students are expected to:

11.1.1: Scientifically document medicinal plants in Papua New Guinea

Activity 1: Mini Project

Title: Documentation of Herbal Plants in PNG

- Students are to follow the documentation format provided in the teacher's resource book with an example given.
- Document traditional/herbal plants in their respective areas
 - Identify a medicinal/herbal plant in their respective areas
 - Research, compile and document the particular plant
 - Keep an account of the documentation
 - Integrate in a system accessible for future reference to help Papua New Guinean citizens

Resources: NIL

(Refer to Chemistry TRB)

Topic 2: Bioactive Compounds in Plants

Performance Indicators;

Students are expected to:

11.1.2a: Define bioactive compounds

11.1.2b: Describe the uses of bioactive compounds in plants

11.1.2c: Discuss different techniques for separation of biomolecules from plant extracts

11.1.2d: Test crude plant extracts against a target

11.1.2e: Observe their effects against the target

Activity 1: Laboratory Session

Title: Extraction of Plant Cyclotides

- Students are to harvest plant tissues and target fish or larvae and do an experiment as stated in the teacher's resource book using the following format:
 - Title
 - Objectives
 - Methodology
 - Results/ Discussions
 - Conclusion
 - Reference

Resources: NIL

(Refer to Chemistry TRB)

Topic 3: Examples of Therapeutic Drugs from plants

Performance Indicators;

Students are expected to:

11.1.3a: Identify therapeutic drugs from nature

11.1.3b: Discuss the importance of the therapeutic drugs.

Activity: Case Study

Title: Case Study of Quinine and Artemisinin (Anti malarials)

- Students are to research on the therapeutic drugs, Quinine and Artemisinin
-
- in and discuss how they were discovered in terms of plant origin and pharmacological indication/use.
 - Plant origin
 - Pharmacological indication/use
- Students are to present their findings and discussions to the class.

Resources: NIL

(Refer to Chemistry TRB)

SAMPLE ASSESSMENTS for MODULE 1

Assessment Methods

- Assignment
- Laboratory Work
- Projects/Researches
- Written Test
- End of term examination

Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
1: Ethnobotany	1.1: Scientifically document medicinal plants in Papua New Guinea 1.2: Discuss traditional plants and therapeutic drugs	1. Medicinal/Herbal Plants	Assignment 1
		2. Bioactive compounds in plants	
		3. Examples of Therapeutic drugs from plants	

Sample Marking Guide

Assessment Task	Module 1: Ethnobotany Topics	Assessment Criteria	Total Marks
		Students will be tested on the ability to;	
Assignment	Medicinal/Herbal Plants	Accurately Document Medicinal/Herbal plants	10
	Bioactive compounds in plants	Accurately state the pharmacological uses of traditional plants	10
	Examples of Therapeutic Drugs from plants	Clearly define therapeutic drug and how they are discovered	10
Total mark			30

THEMA

BIOTECHNOLOGY

INTRODUCTION

Biotechnology combines biology with technology to engineer, modify or upgrade the parts of biological system for industrial and human welfare. It combines various technologies derived from biology, mathematics, physics, chemistry and engineering to modify the biological systems. There are four (4) major areas biotechnology is applied – medical, agricultural, industrial and environmental.

Module 1: DNA Analysis and Proteins

3 Weeks

There are two important techniques which enable development of modern biotechnology. This module will introduce students to these techniques. The first technique is the polymerase chain reaction (PCR) which is the main aspect of chemical engineering and is discussed in Topic 1. Dye chemistry in gene sequencing (chemical engineering) is discussed in Topic 2. Protein expressions which are biological molecules will be covered in Topic 3.

Linking Previous Module (STEM Bridging)

Pre-requisite:

- Module 11.1: Properties and structure of matter
- Module 11. 2: Types of chemical reactions
- Module 11.5: Chemical Bonding

Broad Content Standard

11.1: Discuss DNA Analysis in Polymerase Chain Reaction, Dyes in DNA Sequencing and Proteins

Content Benchmark

Students' ability to demonstrate knowledge and skills in;

- Components, procedures and applications of polymerase chain reaction in health and medicine
- Dyes in DNA sequencing
- Proteins as important macromolecule in living functions.

Topic 1: Polymerase Chain Reaction

Performance Indicators;

Students are expected to:

11.1.1: Describe the components, procedures and applications of polymerase chain reaction (PCR).

Activity 1: Students Task

***Title:* Polymerase Chain Reaction**

- Students are to define PCR
- Students are expected to describe the components of PCR
- Students are to describe the PCR procedures
- Ask students to state and describe briefly some of the useful application of PCR

Resources: NIL

(Refer to Chemistry TRB)

Topic 2: Dyes in DNA Sequencing

Performance Indicators;

Students are expected to:

11.1.2a: Discuss the importance of dye chemistry in gene sequencing

11.1.2b: Discuss different sequencing procedures

Activity: Students Task

Title: Dyes in DNA Sequencing

- Define;
 - a. Dyes
 - b. DNA
 - c. DNA sequencing
 - d. Gene
 - e. Gene sequencing
 - f. Genome
 - g. Genome sequencing
- Describe Sanger DNA sequencing methods;
 - DNA Sequence for Chain Termination PCR
 - Size Separation by Gel Electrophoresis
 - Gel Analysis & Determination of DNA Sequence

Resources: NIL

(Refer to Chemistry TRB)

Topic 3: Proteins

Performance Indicators;

Students are expected to:

11.1.3a: Define proteins

11.1.3b: Identify types of proteins

11.1.3c: Discuss synthesis of proteins

11.1.3d: Discuss case studies of proteins utilized in biotechnology

Activity: Students Task

Title: Proteins

- Students are to;
 - Define proteins
 - Discuss synthesis of proteins
 - Discuss types and functions of proteins
- Discuss Case studies:
 - Cry proteins

Resources: NIL

(Refer to Chemistry TRB)

SAMPLE ASSESSMENTS for MODULE 1

Assessment Methods

- Assignment
- Laboratory Work
- Projects/Researches
- Written Test
- End of term examination

Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
1: DNA Analysis and Proteins	11.1: Discuss DNA Analysis in Polymerase Chain Reaction, Dyes in DNA Sequencing and Proteins	1. Polymerize Chain Reaction	Assignment 1
		2. Dyes in DNA Sequencing	
		3. Proteins	

Sample Marking Guide

Module 1: DNA Analysis and Proteins Topics	Assessment Task	Assessment Criteria	Total Marks
1. Polymerize Chain Reaction	Assignment	Accurately describe PCR	10
2. Dyes in DNA Sequencing		Accurately describe the DNA sequencing	
3. Proteins		Clearly explain what proteins are and their many functions.	10
			10
Total mark			30

THEMA

HEALTH AND

MEDICINE

INTRODUCTION

Medicine is the art, science and practice of caring for a patient and managing the diagnosis, prognosis, prevention, treatment or palliative care of their injury or diseases. Medicine encompasses health care practices evolved to maintain and restore health by the prevention and treatment of illness. The practice of medicine has significantly changed with the advent of molecular biology with its enormous implications in biological sciences (sequencing of human genome), sophisticated new imaging techniques and advances in bioinformatics and information technology. That has sparked an explosion of scientific information that fundamentally influenced and changed the way in which patient care and management has been practiced.

The knowledge gleaned from the science of medicine has already improved and undoubtedly will further improve our understanding of complex disease processes and provide new approaches to disease treatment and prevention. Though the practice of medicine requires a scientific basis, the combination of medical knowledge, intuition, experience and judgment defines the art of medicine.

Furthermore, the recent emergence of SARS-Cov2–2019 (COVID-19) pandemic has shut down the world and caused severe economic downturn globally; it has disrupted the global supply chain. It has exposed the deficiency within the Papua New Guinean health delivery services and their political decision-making processes. This is not the first global outbreak of novel viral diseases. There have been similar outbreaks in the recent past, such as Middle-East Respiratory Syndrome (MERS), Severe Acute Respiratory Syndrome (SARS), Avian Influenza (H1N1 and its variants), African Swine Flu, and so on. It is therefore of paramount importance to educate and prepare the next generation of medical doctors and scientists with sound knowledge in medicine to combat emergence of future pandemics.

Module 1: Biochemistry in Medicine

3 Weeks

Chemistry has many relations to human life. It provides really fundamental knowledge of health and disease. Our bodies are composed of chemical substances and all the numerous functions of the living organisms depend, in one way or another, on chemical processes. Chemical processes enable us to digest food, keep warm and supply us with energy. Even the impressions of our senses, thoughts of our brains, as well as the mode of conveying these through the nerves, are more or less dependent on chemical processes.

Chemistry has made major contributions to health care. The development of new drugs involves chemical analysis and synthesis of new compounds. This is evident in the many recent advertisements made on the large number of new drugs produced.

This module calls attention very briefly to some of the ways in which chemistry may be able to help the theory and practice of medicine.

Linking Previous Module (STEM Bridging)

Pre-requisite:

- Module 11.1: Properties and structure of matter
- Module 11.2: Types of chemical reactions
- Module 11.3: Masses, moles and molarity
- Module 11.4: Acids, bases and salts
- Module 11.5: Chemical Bonding

Broad Content Standard

11.1a: Describe how a medical drug works (mechanism of action)

11.1b: Expound on drug pharmacology

11.1c: Signify diagnostic laboratory results

Content Benchmark

Students' ability to demonstrate knowledge and skills in;

- How a medical drug works
- Drugs pharmacology
- Diagnostic laboratory results

Topic 1: How a Medical Drug Works (Mechanism of Action)

Performance Indicators;

Students are expected to:

11.1.1: Describe the mechanism of action of a medical drug – how it works

Activity: Students Task

Title: Mechanism of Action of a medical drug

1. Students must define a drug
2. Students are to clearly describe how medical drugs are classified.
3. Students to briefly discuss the beneficial purposes of medical drugs
4. Students are to describe how a medical drug works (mechanism of action)
5. Ask students to briefly describe the mechanism of action of Aspirin and Non-steroidal anti-inflammatory drugs (NSAIDs) taking into account the following points:
 - COX 1
 - COX 2
6. Students are to explain how NSAIDs get to the active site?
7. Student to discuss causes of side effects in aspirin and why aspirin causes stomach upset but the newer NSAID drugs do not

Resources: NIL

(Refer to Chemistry TRB)

Topic 2: Drug Pharmacology

Performance Indicators;

Students are expected to;

11.1.2: Describe clearly how drugs work in terms of absorption, distribution, metabolism and elimination and potential toxic effects

Activity 1: Students Task

Title: Drug Pharmacology

- Students are to describe a drug receptor
- Ask the students to define the terms;
 - Efficacy
 - Potency
- Students are to discuss First-Pass Effect
- Students to expound on how drugs cross membranes to show their understanding of drug binding to receptors.
- Students are required to describe how drugs are metabolized and eliminated.

Resources: NIL

(Refer to Chemistry TRB)

Topic 3: Significance of Diagnostic Laboratory Tests

Performance Indicators;

Students are expected to;

11.1.3a: Discuss the types of diagnostic laboratory tests.

11.1.3b: Describe the significance of diagnostic laboratory tests.

Activity 1: Student Task

Title: Significance of diagnostic laboratory tests

- Student are to describe what a Full Blood Count (FBC) is and the reason it is done
- Describe different types of blood cells and state their basic functions.
- Ask students to discuss what they understand about liver function tests.
Ask them to describe briefly the roles of the liver enzymes.
- Ask students to expound briefly on Kidney function tests.

Resources: NIL

(Refer to Chemistry TRB)

SAMPLE ASSESSMENTS for MODULE 1

Assessment Methods

- Assignment
- Laboratory Practicals
- Projects/Researches
- Written Test
- End of term examination

Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
1: Health and Medicine	1.1: Describe how a medical drug works,, drugs pharmacology and diagnostic laboratory results	1. how a medical drug works (mechanism of action)	Laboratory Practical
		2. Drug Pharmacology	
		3. Significance of Diagnostic Laboratory Tests	

Sample Marking Guide

Assessment Task	Module 1: Biochemistry in Medicine Topics	Assessment Criteria	Total Marks
		Students will be tested on the ability to;	
Laboratory Practical	How a medical drug works (mechanism of action)	Precisely describe how a medical drug works.	10
	Drug Pharmacology	Clearly describe drug pharmacology	10
	Significance of Diagnostic Laboratory Tests	Accurately interpret the Diagnostic Tests results	10
Total mark			30

Planning Recording & Reporting Assessment (performance criteria)

Sample Assessment Planning

Assessment Domains	Continuous Assessment per Module				End of Term Exam	Total Marks	% Weight of Domain
	Biodiversity		Biotechnology	Health & Medicine			
	M1	M2	M1	M1			
Theory Knowledge & Comprehension	Research Project (25)	Research Project (25)	Test (25)	Test (25)		100	80
Practical & Experimental Skills							
Application of knowledge							
Total Mark	25	25	25	25			
% of contribution of Each Module	20%	20%	20%	20%	20%	100%	100

Outlined in the table above is an overall assessment per module for the for the three Thema (Biodiversity, Biotechnology, Health and Medicine) of term 2 of Grade 11. It covers the key domains of a holistic approach to learning and understanding key concepts in biology. The teacher is given the autonomy of setting whatever assessment task that is necessary for a given module or individual topic, however is required to comply with the assessment guidelines set above for uniformity throughout the country.

Sample Assessment Recording

No.	Names	Assessment Task for Thema Biodiversity		Total Score (/50)	Total Percentage Weight (%)	Achievement Level [VHA, HA, SA,LA]
		RP1 (/25)	RP1 (25)			
1	Komun Bomai	25	25	50	100%	VHA
2	Drima Dane	20	20	40	80%	HA
3	Thokma Diur	15	15	30	60%	SA

Grading Criteria

Percentage Range	Achievement Level
85 -100	Very High Achievement (VHA)
65-84	High Achievement (HA)
50-64	Satisfactory Achievement (SA)
25-49	Low Achievement (LA)
0 - 24	Below Minimum Achievement (BMA)

Sample Reporting: Test				
Performance Criteria	Accurately answers All the Multiple Choice Questions (10 marks)	Precisely answers all the Short Answers (10 marks)	Correctly labels the diagram (5 marks)	25 marks
Very High Achievement	Accurately interprets the questions and selects the correct answers (9-10)	Accurately interpret and precisely answers the questions (9-10)	Accurately labels the diagram (4-5)	21–25
High Achievement	Partially interprets the questions and selects the correct answers (7-8)	Partially interprets and answers the questions (7-8)	Partially labels the diagram (3)	16–20
Satisfactory Achievement	Fairly interprets the questions and selects the correct answers (5-6)	Fairly interprets the questions and answers (5-6)	Impartially labels the diagram (2)	9–13
Low Achievement	Poorly interprets the questions and selects the correct answers (3-4)	Poorly interprets the questions and answers (3-4)	Poor understanding of the diagram (1)	(6–8)
Below minimum Achievement	Needs improvement in understanding and interpreting questions and selecting the correct answers (0-4)	Needs improvement in understanding and interpreting questions and providing the correct answers (0-4)	Does not understand the diagram and cannot label correctly (0)	(0-5)
Sample Reporting: Laboratory Practical 20 marks				
Performance Criteria	Very High Achievement (17–20)	High Achievement (14–17 Marks)	Satisfactory Achievement (9–13 Marks)	Low Achievement (0–8 Marks)
Appropriate	Correctly and	Clearly state	Clearly state	Inappropriate

title (2 marks)	clearly state the required title (2)	the title (1-2)	the title (1-2)	title (01)
Suitable aim (2 marks)	Aims clearly and correctly written (2)	Aims clearly stated (1-2)	Aims clearly stated (1-2)	Irrelevant aims (0-1)
Logical procedures (6 marks)	Correct procedures followed and equipment used (6)	Procedures followed and appropriate equipment used (4-6)	Procedures and the use of equipment reasonably followed (1-4)	Inconsistent procedures and use of equipment (0-1)
Detailed analysis and explanation (6 marks)	Scientific analysis with detailed explanation and logical presentation (6)	Good analysis and explanation (5)	Fair analysis and explanation (3-4)	Poor analysis with minimum explanation (0-2)
Appropriate conclusion (4 marks)	Scientific and reliable conclusion (4)	Reliable and well written conclusion (3)	Conclusion appropriate (2)	Inappropriate conclusion Lacks scientific understanding (0-1)
20 marks	22/20 VHA	14/20 HA	10/20 SA	7/20 LA

REFERENCE

1. J. Sharwood, N.Taylor, R. Kronenberg, M. Mc Laughlin, T. Sweeney, D. Corrigan and S. Rodrigues (2000) Chemistry, Nelson Thomson Learning, Australia
2. Taylor, M. Derbogosian, Wan Ng, A. Stubbs, R.Stokes, S.Bowen, S.Raphael, J. Moloney, (2007) Chemistry 1 John Wiley & Sons Wiley Ltd, Australia,
3. <https://www.genome.gov/about-genomics/fact-sheets/Polymerase-Chain-Reaction-Fact-Sheet>, [accessed on 16/02/2021;1.35pm]
4. <https://www.khanacademy.org/science/ap-biology/gene-expression-and-regulation/biotechnology/a/polymerase-chain-reaction-pcr>, [accessed on 10/03/2021;1.40pm]
5. https://en.wikipedia.org/wiki/Polymerase_chain_reaction, [accessed on 20/03/2021;2pm]
6. Carter W, Bowen J. Pathology: Blood tests. In: The Macquarie Home Guide to Health and Medicine. McMahon's Point, NSW: Macquarie Library; 1991; 201-2.
7. Isbister JP. Clinical presentations of haematological disease. *Medicine Today*. 2004; 5(10): 27-38.
8. Isbister JP. Investigating and treating anaemia. *Current Therapeutics*. 2000; 41(10): 39-44.
9. McFerran T. Oxford Minidictionary for Nurses [4th Ed]. 1998. Oxford: Oxford University Press.
10. Smellie W, Wilson D, McNulty C, Galloway M, Spickett G, Finnigan D, *et al*. Best practice in primary care pathology: Review 3. *Journal of Clinical Pathology*. 2006; 59(8): 781-9.
11. Smellie W, Forth J, Smart S, Galloway M, Irving W, Bareford D, *et al*. Best practice in primary care pathology: Review 7. *Journal of Clinical Pathology*. 2007; 60(5): 458-65.
12. Warburton P. Investigating thrombocytosis. *Medicine Today*. 2005; 6(1): 32-9.
13. WA School of Pathology Training Manual. WA School of Pathology. 2007. Wangara.

GLOSSARY

- **Cholinergic** – relating to nerve cells that use acetylcholine as their neurotransmitters.
- **Antimalarial** – a chemotherapeutic agent that inhibits or destroys malarial parasites.
- **Absorption** – the taking in, incorporation or reception of gases, liquids,, light or heat.
- **Acidity** – the acid content of a fluid
- **Adrenergic** – relating to nerve cells or fibres of the autonomic nervous system that use norepinephrine as their neurotransmitter
- **Agonist** – a drug capable of combining with receptors to initiate drug actions.
- **Albumin** – a type of simple protein, varieties of which are widely distributed throughout the tissues and fluids of plants and animals.
- **Analgesic** – a compound capable of producing analgesia, on that relieves pain by altering the perception of nociceptive stimuli without producing anesthesia or loss of consciousness.
- **Antagonists** - drugs that interact selectively with receptors but do not lead to an observed effect.
- **Anthropology** – the branch of science concerned with the origin and development of humans in all their physical, social and cultural relationship
- **Bactericidal** – causing the death of bacteria.
- **Basophil** – a type of white blood cell
- **Bilirubin** – a yellow bile pigment
- **Diuretic** – promoting excretion of urine
- **Enzyme** – a macromolecule that acts as a catalyst to induce chemical changes in other substances, while itself remaining unchanged by the process.
- **Eosinophil** – a type of white blood cell responsible for allergies and asthma
- **Neutrophil** – a type of white blood cell responsible for most of the infectious diseases.
- **Heamoglobin** – a type of protein found in the blood that is responsible in the transport of oxygen throughout the body tissues.
- **Platelets** – a constituent of the blood that is responsible for the clotting
- **Receptor** – a structural protein molecule on the cell surface or within the cytoplasm that binds to a specific factor, such as drug, hormone, antigen or neurotransmitter